

```

clc; clear; close all;

%% --- Input Parameters ---
fp = 1000;      % High-pass cutoff frequency (Hz)
fs = 8000;      % Sampling frequency (Hz)
N = 50;         % Filter order
wc = fp/(fs/2); % Normalized cutoff (0-1)

%% --- FIR High Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'high', hamming(N+1));
disp('--- High-Pass Coefficients (Hamming Window) ---');

--- High-Pass Coefficients (Hamming Window) ---

```

```

disp(b_hamm');

```

```

-0.0007
  0
  0.0009
  0.0016
  0.0015
  0
-0.0024
-0.0044
-0.0039
  0
  0.0060
  0.0103
  0.0088
  0
-0.0128
-0.0217
-0.0184
  0
  0.0268
  0.0464
  0.0410
  0
-0.0725
-0.1567
-0.2240
  0.7493
-0.2240
-0.1567
-0.0725
  0
  0.0410
  0.0464
  0.0268
  0
-0.0184
-0.0217
-0.0128
  0
  0.0088
  0.0103
  0.0060
  0
-0.0039
-0.0044

```

```
-0.0024  
0  
0.0015  
0.0016  
0.0009  
0  
-0.0007
```

```
%% --- Frequency Response Plots ---  
figure;  
freqz(b_hamm,1);  
title('High-pass FIR using Hamming Window');
```

